

Research

AI versus human effectiveness in essay evaluation

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Abstract

The evaluation of student essay corrections has become a focal point in understanding the evolving role of Artificial Intelligence (AI) in education. This study aims to assess the accuracy, efficiency, and cost-effectiveness of ChatGPT's essay correction compared to human correction, with a primary focus on identifying and rectifying grammatical errors, spelling, sentence structure, punctuation, coherence, relevance, essay structure, and clarity. The research involves collecting essays from 100 randomly selected university students, covering diverse themes, with anonymity maintained and no prior corrections by humans or AI. An analysis sheet, outlining linguistic and informational elements for evaluation, serves as a benchmark for assessing the quality of corrections made by ChatGPT and humans. The study reveals that ChatGPT excels in fundamental language mechanics, demonstrating superior performance in areas like grammar, spelling, sentence structure, relevance, and supporting evidence. However, thematic consistency remains an area where human evaluators outperform the AI. The findings emphasize the potential for a balanced approach, leveraging both human and AI strengths, for a comprehensive and effective essay correction process.

Keywords Higher education · Essay correction · AI vs human evaluation · Effectiveness

1 Introduction

Automated Essay Scoring (AES) tools have evolved significantly since their appearance, initially propelled by Ellis Page's Project Essay Grader (PEG) in 1966, which pioneered the use of statistical methods to correlate textual features with human rater preferences [1]. Despite early promise, AES faced limitations in capturing the subtle aspects of content evaluation that human graders excel in, such as understanding context, creativity, and coherence [2]. These limitations have been a subject of criticism within the educational assessment community, where concerns over the reliability and fairness of automated grading persist [3].

The re-emergence of interest in AES in the late 1990s and early 2000s was fueled by advancements in natural language processing (NLP) and machine learning techniques, which offered new possibilities for improving the accuracy and reliability of automated grading systems [3]. Despite these advancements, current AES tools still struggle with certain aspects of content evaluation, often demonstrating disparities in scoring when compared to human graders, particularly in assessing complex, argumentative, or creative writing [3].

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The need for further research in this domain is emphasized by the ongoing quest to develop AES tools that can effectively emulate human judgment while leveraging the scalability and efficiency offered by AI technologies [4]. Recent developments in AI, particularly with the emergence of large-scale language models like ChatGPT, present a promising avenue for addressing these challenges. ChatGPT, developed by OpenAI, represents a significant advancement in NLP, capable of understanding and generating human-like text with a high degree of coherence and contextuality [5].

Hence this study attempts to explore the potential of ChatGPT in enhancing essay evaluation by conducting a comparative analysis between its performance and that of human graders across various evaluation metrics, including grammar, coherence, thematic consistency, and overall quality. By empirically evaluating ChatGPT's capabilities alongside human assessments, this research seeks to provide insights into the strengths and limitations of AI-driven grading systems in educational settings. Such insights are crucial for contributing to the development of more effective and reliable automated grading tools, thereby advancing the field of educational technology and assessment practices [6].

2 Literature review

The following literature review critically examines Automated Essay Scoring (AES) tools like PEG, e-rater, and IntelliMetric™, tracing their evolution and methodologies from early statistical approaches to advanced natural language processing models. These tools have significantly influenced educational assessment by automating essay evaluation based on predefined criteria. The reviewed studies will evaluate the effectiveness and reliability of these tools in comparison to human graders and address criticisms of their capability to assess creativity and contextual complexities in writing. This review aims to contextualize findings within the dynamic field of AES, setting the tone for an exploration of ChatGPT's potential contributions. As an advanced AI language model capable of contextual understanding and generating coherent text, ChatGPT represents a promising advancement in addressing the limitations of current AES tools. Evaluating its role in essay assessment will provide insights into the future of AI-driven educational technology and assessment practices.

2.1 Automated essay scoring tools

The field of Automated Essay Scoring (AES) has a history spanning 40 years, with Ellis Page recognized as its pioneer. Page's Project Essay Grader (PEG) in 1966 was a significant early attempt, employing statistical methods to identify textual features correlating with human raters' preferences. Despite initial promise, AES tools faced limited popularity until the late 1990s. Advances like the Writer's Workbench and Writer's Helper in the 1980s shifted focus to feedback rather than scoring. The 1990s witnessed attempts to correlate word choice with writing quality, leading to more advanced designs. Major AES devices like Intelligent Essay Assessor, e-rater, and IntelliMetric™ emerged in the late 1990s, utilizing natural language processing and statistical technologies [7–9]. However, current AES tools still face challenges in content evaluation compared to human raters, emphasizing the need for ongoing advancements in AI and NLP research. Major examples from past studies will be reviewed below:

To assess the efficiency of Project Essay Grade (PEG) in evaluating specific writing traits, Page, Poggio, and Keith [1] conducted a study using a sample of 495 essays written by 12th graders for the 1988 National Assessment of Educational Progress (NAEP) writing assessment. Employing the Spearman-Brown prophecy test, they evaluated PEG's predictive accuracy for average scores from eight raters in comparison to the predictive performance of two, three, and four human raters. The findings indicated that PEG outperformed the predictive rates of two human raters across all trait rating scores and holistic scores. However, in comparison to four-rater prediction rates, PEG's predictive rates were lower, particularly for holistic scores and two specific traits: style and mechanics [1].

A different investigation into the credibility of PEG in assessing both specific essay traits and overall quality was carried out by Shermis, Koch, Page, Keith, and Harrington [3]. This study employed Confirmatory Factor Analysis (CFA) to assess PEG's validity by comparing its scores to those assigned by all potential pairs of six human raters. To ensure distinct pairs, five separate analyses were conducted. The outcomes revealed that the standardized pattern coefficient for human pairs ranged from 0.81 to 0.89, with a median coefficient of 0.86. In contrast, PEG's coefficients ranged from 0.88 to 0.89, with a median coefficient of 0.89. These results implied that the computer ratings of essays were to a certain extent as valid as pairs of human judges [3].

Besides Project Essay Grade (PEG), other studies have been conducted on an Automated Essay Scoring (AES) tool called e-rater. Within this context, Burstein, Kukich, Wolff, Lu, and Chodorow [2] conducted a study assessing the validity

of the e-rater tool by scoring 500 Graduate Management Admissions Test (GMAT) essays and 200 Test of Written English (TWE) essays. Correlation analyses demonstrated that e-rater had comparable correlation rates with human raters. Notably, the study examined discrepancies, revealing higher rates between e-rater and human raters at score levels 5 and 6 compared to discrepancies between two human raters. The findings suggested strong support for the validity of e-rater in essay scoring.

To investigate the limits of the AES tool e-rater, Powers, Burstein, Fowles, Chodorow, and Kukich [10] invited writing experts and critics of AES to manipulate responses to the Graduate Record Examinations (GRE) writing prompts to trick e-rater. The study found that 67% of the writing samples were correctly predicted by e-rater, with 17% in the wrong direction and 17% having identical e-rater ratings despite predicted differences. The researchers recommended using e-rater alongside human raters and implementing filters to identify specific essay characteristics, emphasizing the need for a combined approach to enhance accuracy and address potential challenges in automated essay scoring [8].

More studies have been done on e-rater such as the one made by Attali and Burstein [4] who investigate the application of the e-rater V.2 system for automated essay scoring, that was employed by the Educational Testing Service for automated essay scoring since 1999. According to their study, E-rater V.2 distinguishes itself from other scoring systems with its innovative features, including a concise and meaningful set of scoring criteria applicable to all prompts, a unified scoring model and standards, and transparent and flexible modeling procedures grounded in expert judgment. Their paper outlines the characteristics of this new system and provides evidence supporting the validity and reliability of its scoring outcomes.

E-rater version 2.0 has been again investigated along with Criterion in a study by Burstein and Chodorow [10] in which they introduce the Criterion online writing service, showcasing its application of artificial intelligence in written content assessment. The authors highlight two main components: Critique Writing Analysis Tools, adept at detecting errors and identifying elements in essays, and e-rater version 2.0, an automated essay scoring system. Both provide personalized feedback to enhance student writing skills under the guidance of a classroom teacher. These tools, employing natural language processing and machine learning, surpass basic algorithms, occasionally aligning with human judges' evaluations as consistently as two human judges agree with each other.

In a related study, Ramineni et al. [11] employed the e-rater to analyze 4000 Test of English as a Foreign Language (TOEFL) essays, demonstrating a strong positive correlation and substantial agreement between automated and human scorings for both independent and integrated tasks. Similarly, Evanini et al. (2015) investigated 3385 secondary school students' TOEFL Junior Test responses, revealing a strong positive correlation and substantial agreement across all test items between automated and human scoring.

Exploring further tools in automated essay scoring (AES), Dikli [12] provides methods to enhance the reliability of AES systems, emphasizing the calibration with diverse essay samples. Dikli discusses the emergence of student-based tools like MY Access!® and CriterionSM, originally designed for large-scale essay scoring but currently utilized as writing tools in educational settings. The article anticipates the potential transformation of AES systems into a public utility for broader educational access. While acknowledging the increasing demand for AES in writing assessments, the author emphasizes their role as supportive tools for teachers rather than replacements. The article also suggests integrating AES as a second or third rater to validate human scoring and represent a diverse collection of human judgments in large-scale assessments. Overall, Dikli's work [12] offers insights into the evolving landscape of AES, its applications, and considerations for future developments.

Another automated essay scoring (AES) tool called IntelliMetric™ is investigated; initial research primarily focuses on validating its effectiveness. Over 120 studies since 1996 have predominantly used correlational designs to compare scores assigned by IntelliMetric™ with those given by human raters. The majority of these studies reported high agreement rates and correlational coefficients, with Rudner, Garcia, and Welch [13] noting a correlational coefficient rate as high as 0.83. While this body of research supports the validity of IntelliMetric™. The limited number of studies employing comparative research methodologies indicated that there were no statistically significant differences between the scores assigned by Automated Essay Scoring (AES) systems and those assigned by human raters.

Within the context of comparative methodologies between automated and human scoring conducted on IntelliMetric™, Nivens-Bower [14] conducted a comparative study at two New England community colleges, assessing IntelliMetric™ against human raters for the WritePlacer Plus test. The results showed no significant differences in mean scores, leading to the conclusion that IntelliMetric™ produced results consistent with faculty scores. Vantage Learning (2003) applied IntelliMetric™ to grade instructional literary analysis essays, reporting no significant difference between the mean scores assigned by experts and IntelliMetric™, along with high agreement and correlation coefficient rates. Despite the limited number of comparative studies, they consistently suggest alignment between IntelliMetric™ and human raters.

Comparing automated essay scoring (AES) to human scoring, Wang and Brown's [15] study assesses the validity of (AES) by comparing group mean scores assigned by AES with those from human raters. The study, conducted at a Hispanic-serving institution in South Texas with 107 participants, examines the performance of IntelliMetric™ compared to human raters on WritePlacer Plus and THEA writing tests. Statistical analyses reveal significant differences, with IntelliMetric™ assigning higher mean scores than human raters on WritePlacer Plus and surpassing THEA mean scores. The findings challenge prior studies reporting non-significant mean score differences between AES and human scoring. The authors recommend future research focusing on surface features and mean score disparities, suggesting a comprehensive approach to assessment programs for accurate placement and understanding students' learning needs.

Within the same objective of reducing resource-intensity of human scoring and achieving more consistent scoring results than human raters, Wahlen et al. [16] conducted a comparative study where they analyzed scores given by both the automated scoring system ESCRITO and human raters for learners' responses to open-ended tasks. The authors observed a significant agreement between automated and human ratings, affirming that the automated scoring system evaluates the same aspects as human rating. This assertion is grounded in the fact that both computers and humans produced convergent rating outcomes (p.8). Furthermore, several researchers have emphasized the benefits of incorporating computer-assisted grading applications, noting that they do not negatively impact student attitudes regarding the helpfulness of feedback, satisfaction with feedback speed, or satisfaction with the feedback delivery method.

This study assesses the accuracy of PaperRater, an automated essay scoring tool, in evaluating written performances of EFL learners compared to human raters. Unlike previous research, this study reveals marked disparities in scores, with PaperRater assigning notably higher scores. Despite the cost-effectiveness of combining automated and human scoring, the study indicates a lack of consistency and reliability compared to human assessment alone. Nonetheless, the study recognizes the value of computerized scoring systems in education for their role in enhancing the learning process, particularly through instant feedback contributing to the improvement of EFL learners' writing skills.

2.2 Criticism of automated essay scoring (AES)

The above past studies on Automated Essay Scoring (AES) tools such as PEG, e-rater, Criterion, IntelliMetric, ESCRITO, and others have provided valuable insights into their functionality and effectiveness. However, these studies also highlight several critical limitations and challenges inherent in current AES systems. Despite advancements in natural language processing and machine learning, AES tools often struggle to capture and evaluate the creativity and originality present in students' writing. These systems primarily rely on predefined scoring criteria and patterns learned from training data, which may overlook innovative expressions and unique writing styles that do not conform to standardized norms. This limitation undermines the ability of AES tools to fully recognize and reward creative thinking and specificities of argumentation in essays Elliot's [17].

This is also confirmed by Broad's [18] and Perelman's [19] criticism stating that AES have limited ability to assess creativity, critical thinking, and context in student writing. He argues that AES overly focuses on surface-level features like grammar and vocabulary, neglecting deeper aspects of writing quality such as coherence and argumentation. Perelman [19] also highlights concerns about AES's ineffectiveness in detecting plagiarism and its potential impact on reducing pedagogical emphasis on critical writing skills. He advocates for a balanced approach that integrates automated scoring with human judgment to ensure a more holistic evaluation of students' writing abilities.

Shermis and Burstein [20] further, in their work on automated essay scoring (AES), highlight a significant critique regarding the systems' ability to comprehend the subtle context of essays. Unlike human graders who can interpret meaning from subtle textual cues and understand the situational context of arguments, AES systems primarily assess essays based on surface-level features like grammar, coherence, and structural adherence. This reliance on superficial criteria can lead to misinterpretations or oversights regarding the author's intent, especially in essays discussing abstract topics where contextual understanding is essential for accurate assessment. Consequently, AES tools may struggle to provide accurate feedback and scores when confronted with ambiguous or unconventional writing styles, thereby limiting their effectiveness in capturing the full breadth of students' writing capabilities.

Moreover, the unidirectional nature of AES tools, where they assess essays without engaging in interactive dialogue or adapting to evolving discourse, limits their ability to provide meaningful, context-aware evaluations. This one-size-fits-all approach contrasts with the adaptive capabilities of Intelligent Automation (IA) Language Models like ChatGPT, which can dynamically interpret and respond to varying styles and contexts in real-time. Therefore, addressing these limitations is crucial for advancing AES technologies towards more reliable and comprehensive essay evaluation systems that align with the evolving educational needs and expectations of today's learners and educators.

2.3 AI language model: ChatGPT in essay evaluation

In the context of essay evaluation, intelligent Automation (IA) Language Models, exemplified by ChatGPT, offer a transformative alternative to Automated Essay Scoring (AES). These models go beyond the limitations of AES by excelling in contextual understanding. ChatGPT, for instance, showcases a remarkable capacity to comprehend context, capturing the linguistic complexities and recognizing subtle expressions. This capacity is expected to provide a more refined and accurate evaluation, particularly in assessing essays on diverse and complex topics.

The potential advantages of employing AI language models in essay evaluation are manifold. For example, ChatGPT can provide a more dynamic and interactive assessment, considering the individuality of each student's writing. Its contextual understanding allows for a more comprehensive evaluation of creativity, originality, and coherence. Additionally, the continuous learning capabilities of AI language models allow them to stay updated with evolving language trends and adapt to changes in communication styles. Overall, the integration of advanced AI language models in essay evaluation holds the potential to offer a more holistic, accurate, and adaptive approach compared to conventional scoring methods.

Latest studies have confirmed the potential of AI language model ChatGPT in bringing about significant transformations in the context of foreign language research, teaching, and learning. A recent study by Mizumoto and Eguchi [21] explores leveraging this technology for automated essay scoring (AES) and assesses its dependability and precision. Specifically, the authors employ the GPT-3 text-davinci-003 model to automatically evaluate all 12,100 essays found in the Educational Testing Service (ETS) Corpus of Non-Native Written English (TOEFL 11), comparing these assessments with established benchmark levels. The study also tackles the impact of linguistic features on AES using GPT. The findings indicated that AES with GPT attains a discernible level of accuracy and reliability, offering valuable support for human evaluators. Moreover, the analysis demonstrated that incorporating linguistic features can enhance the precision of scoring. These results suggest that AI language models, including ChatGPT, can be effectively employed as AES tools, potentially revolutionizing approaches to writing evaluation and feedback in both research and practical applications.

Within the context of essay evaluation, another research by Steiss et al. [22] explores ChatGPT's ability to offer formative feedback on student compositions, comparing it with human evaluators across 200 essays. Criteria include alignment with standards, clarity, accuracy, emphasis on crucial elements, and maintaining a supportive tone. Results indicate that human evaluators excel in delivering high-quality feedback across all categories except for criteria-based feedback. Despite the ease of generating feedback with ChatGPT and its generally sound quality, practical distinctions between human evaluators and ChatGPT are not significant. Feedback quality does not exhibit variations based on language proficiency for both human evaluators and AI, but distinctions are observed between AI and human feedback based on essay quality. The study discusses implications for the use of generative AI as an educational tool.

In the context of evaluating students' assignments, Dai et al.'s study [23] explores the use of ChatGPT, as a pre-trained language model, in providing feedback to students. Recognizing the effectiveness of feedback in enhancing learning, the researchers aim to leverage ChatGPT's capabilities to generate natural-sounding and detailed responses. The investigation assesses the readability of ChatGPT-generated feedback, compares its assessment of student assignments with that of an instructor, and employs a theoretical feedback framework for evaluation. Results indicate that ChatGPT can produce more detailed and coherent feedback than human instructors, exhibits high agreement in evaluating assignment content, and contributes to students' skill development by providing feedback on the task completion process.

In addition to its evaluative capabilities, ChatGPT has been noted for its potential to promote reflective and critical thinking skills among students. The ability of ChatGPT and similar tools to stimulate creativity, problem-solving skills, and conceptual comprehension positions them as valuable "objects-to-think-with" in educational practices [24]. Integrating such AI language models into educational assessments offers the promise of enhancing deeper engagement and learning outcomes through personalized and insightful feedback [24]. Additionally, Ghafar [25] discusses the potential of ChatGPT in enhancing the teaching and evaluation of second and foreign languages, highlighting its role in customized education and formative evaluation. This perspective aligns with the broader educational applications of ChatGPT, suggesting its utility extends beyond traditional essay scoring to include personalized instructional strategies and enhanced language learning outcomes [25].

Incorporating ChatGPT into educational assessments not only addresses current challenges in essay evaluation but also opens new avenues for personalized and effective learning environments. Continued research and implementation will be crucial in fully leveraging the potential of AI language models like ChatGPT to transform educational practices and improve student outcomes.

Generally, the above section has discussed the potential of AI language models, particularly ChatGPT, in the context of essay evaluation, emphasizing their advantages over traditional tools. The presented studies, however, are notably scarce, underscoring the need for further exploration, particularly in the context of ChatGPT’s accuracy and efficiency in essays correction. Further studies can address the strengths and potential limitations of using ChatGPT for evaluating a diverse range of students’ essays. Such research is essential to establish the reliability, validity, and ethical considerations of integrating AI language models into educational assessment practices.

3 Methods

3.1 Study design and participants

This study employs a comparative design to evaluate the effectiveness of AI (ChatGPT) and human experts in essay correction. A sample of 100 essays was collected from students at a university institute with a total enrollment of nearly 900 students. The students were randomly selected to ensure diversity in essay themes and to avoid any selection bias. The essays collected were anonymized to ensure that neither the AI nor human evaluators had access to identifying information about the students. The information concerning the students is explained in Table 1.

3.2 Essay collection process

The essays were gathered in a controlled setting to maintain consistency. Students were invited to a computer room designated for this study, where they composed their essays. Once completed, the essays were submitted via a dedicated email address created specifically for this research. This process ensured uniformity in the writing environment and in the submission method.

3.3 Development of evaluation criteria

Before initiating the correction process, an evaluation criteria sheet was developed. This sheet outlined specific linguistic and informational elements essential for assessing the essays’ quality. The criteria were designed to evaluate content, organization, coherence, and overall quality of the essays. The development of these criteria involved a group of experienced educators, whose expertise is reflected in the elements selected. The evaluation criteria are detailed in Fig. 1.

3.4 Evaluation criteria and expert panel

The evaluation criteria sheet encompasses elements for assessing both the quality of student writing and the quality of feedback provided by ChatGPT and human raters. The criteria for assessing the quality of student writing include:

- Grammar: Proper use of tenses, verb agreement, and sentence construction.
- Spelling: Accurate spelling throughout the essay.
- Sentence Structure: Clear and varied sentence structures.
- Punctuation: Appropriate use of punctuation marks.
- Coherence: Logical flow and transitions.

Table 1 Participants of the study

Attribute	Details
Student Gender	62 female, 48 male
Writing Fields	42 students chose to write essays on management topics 30 students chose to write essays on economics topics 28 students chose to write essays on literature topics
Student Age	All students are between 19 and 24 years old

Fig. 1 Analysis sheet

Notation	1	2	3	4	5
Grammar	☐	☐	☐	☐	☐
Spelling	☐	☐	☐	☐	☐
Sentence Structure	☐	☐	☐	☐	☐
Punctuation	☐	☐	☐	☐	☐
Coherence	☐	☐	☐	☐	☐
Relevance	☐	☐	☐	☐	☐
Essay Structure	☐	☐	☐	☐	☐
Clarity	☐	☐	☐	☐	☐
Thematic Consistency	☐	☐	☐	☐	☐
Supporting Evidence	☐	☐	☐	☐	☐

- Relevance: Content alignment with the essay prompt.
- Essay Structure: Clarity and organization of introduction, body, and conclusion.
- Clarity: Concise expression and avoidance of ambiguity.
- Thematic Consistency: Maintenance of the essay's overall theme.
- Supporting Evidence: Adequate use of examples and references.

Before presenting the results of our evaluation of the feedback, it is important to note that the elements listed on the developed analysis sheet consist of the following elements in Table 2.

The evaluation criteria sheet was developed by a panel of 12 experts. This panel consisted of educators with significant expertise in essay evaluation. Each member holds a PhD and has at least 10 years of teaching experience in the humanities and social sciences. The qualifications and expertise of the panel ensured that the evaluation criteria were both rigorous and relevant. The details of the expert panel are provided in Table 3.

Table 2 Details of the evaluation sheet

Criteria	Evaluation focus
Grammar	1. Correct identification and resolution of grammatical errors (Feedback quality) 2. Proper use of tenses, verb agreement, and sentence construction (Student writing quality)
Spelling	Accurate spelling of words throughout the essay (Student writing quality)
Sentence Structure	1. Clear and varied sentence structures (Student writing quality) 2. Proper use of complex and compound sentences (Student writing quality)
Punctuation	Appropriate use of punctuation marks (commas, periods, semicolons, etc.) (Student writing quality)
Coherence	Logical flow of ideas and smooth transitions between paragraphs (Student writing quality)
Relevance	Relevance of content to the essay prompt or topic; Avoidance of irrelevant information (Student writing quality)
Essay Structure	1. Introduction: Clarity, thesis statement, and context establishment (Student writing quality) 2. Body: Clear organization of main ideas and supporting details (Student writing quality) 3. Conclusion: Effective summarization and closure (Student writing quality)
Clarity	Clear and concise expression of ideas; Avoidance of ambiguity and vagueness (Student writing quality)
Thematic Consistency	How well the information maintains consistency with the overall theme or subject matter (Student writing quality)
Supporting Evidence	Are there sufficient and appropriate examples, facts, or references to support the presented information (Student writing quality)

3.5 Essay correction procedure

The 100 essays were randomly assigned to two groups for correction: one group was evaluated by ChatGPT, and the other by human experts. To ensure fairness and avoid bias, the corrections were made independently, with each evaluator group unaware of the method used by the other group. The human experts were professors from the English department, all of whom were qualified and experienced in essay correction.

3.5.1 Scoring examples

Tables 4 and 5 illustrate examples of the scoring process for corrections made by ChatGPT and human experts, respectively. These examples provide details how the evaluation criteria were applied to assess the quality of the corrections.

3.6 Secondary evaluation

To validate the accuracy and consistency of the corrections, a secondary evaluation was conducted. This involved five full professors, each with nearly 25 years of experience in teaching and mentoring. These evaluators, all male and aged between 50 and 55, reviewed the corrections made by both ChatGPT and human experts. The secondary evaluation ensured that the assessment of corrections was comprehensive and unbiased.

Table 3 Information about the group of experts

Attribute	Details
Number of Experts	12
Gender of Experts	7 male, 5 female
Specialty	Humanities and Social Sciences
Degree	PhD
Years of Teaching Experience	10 years

Table 4 Example of the scoring of ChatGPT correction

Students	Grammar	Spelling	Sentence structure	Punctuation	Coherence	Relevance	Essay structure	Clarity	Thematic consistency	Supporting evidence
Student #1	5	5	5	5	5	5	5	5	5	5
Student #2	5	5	5	5	5	5	5	5	5	5
...	...	...	...	...	...	...	...	...	...	...
Average	5	5	5	5	5	5	5	5	5	5

Table 5 Example of the scoring of human correction

Students	Grammar	Spelling	Sentence Structure	Punctuation	Coherence	Relevance	Essay Structure	Clarity	Thematic consistency	Supporting evidence
Student #1	5	5	5	5	5	5	5	5	5	5
Student #2	3	3	3	3	3	3	3	3	3	3
...	...	...	...	...	...	...	...	...	...	...
Average	4	4	4	4	4	4	4	4	4	4

3.7 Student feedback

In addition to quantitative analysis, qualitative data were collected through interviews with 10 students whose essays had been corrected. The purpose of these interviews was to gather students' opinions on the quality and usefulness of the corrections provided by ChatGPT and human experts. The feedback aimed to assess students' satisfaction and perceptions of the effectiveness of the feedback received.

3.8 Data analysis

The study employed both quantitative and qualitative methods for data analysis. Quantitative data were analyzed using statistical techniques such as paired t-tests and ANOVA to compare the effectiveness of AI and human corrections. Qualitative feedback from student interviews was analyzed thematically to identify common themes and insights related to the perceived quality of the corrections.

3.9 Ethical considerations

Ethical approval for the study was obtained from the university's ethics committee. All participants provided informed consent before participating in the study, and their anonymity was maintained throughout the research process. Ethical guidelines were strictly followed to ensure the integrity and credibility of the research.

3.10 Limitations and delimitations

Limitations: The study's limitations include potential biases in the evaluation process despite blinding, and the relatively small sample size of 100 essays, which may not fully represent all possible essay types and topics. The generalizability of the findings may also be limited by the specific context of the university institute and the evaluation criteria used.

Delimitations: The study focused on essays from a single university institute and utilized specific evaluation criteria, which may not be applicable to other educational contexts or languages. The research scope was limited to English essays, which may affect the applicability of the findings to other languages or disciplines.

4 Results and discussion

The study evaluated essay corrections provided by both human experts and ChatGPT using various criteria. The average ratings of the corrections are summarized in Table 6.

- Grammar and Spelling:**

The AI consistently outperformed humans in grammar and spelling corrections, showcasing its strength in handling language mechanics. This suggests that AI technologies, such as ChatGPT, are highly effective in addressing fundamental language structure and accuracy.
- Sentence Structure and Punctuation:**

Like grammar and spelling, the AI excelled in correcting sentence structures and punctuation. This is indicative of the AI's ability to understand and implement complex sentence constructions and punctuation rules accurately.
- Coherence and Relevance:**

The AI demonstrated improvement in coherence and excelled in ensuring relevance. This suggests advancements in AI language models in understanding context and delivering content that aligns closely with the essay prompt or topic.
- Essay Structure and Clarity:**

Both humans and AI performed comparably well in essay structure and clarity. This implies that, in these aspects, the AI is on par with human evaluators, providing consistent and clear essay structures.
- Thematic Consistency:**

Humans outperformed the AI in maintaining thematic consistency. This might be attributed to the nuanced understanding and contextual interpretation that humans possess, allowing them to better ensure the seamless flow of ideas throughout an essay.
- Supporting Evidence:**

The AI significantly outshone humans in providing supporting evidence. This highlights the AI's capacity to incorporate relevant examples and facts to bolster the essay's arguments, potentially adding depth to the overall content.

This detailed analysis indicates that the AI generally performed exceptionally well, especially in areas such as grammar, spelling, sentence structure, relevance, and providing supporting evidence. However, there are instances where human corrections, such as in thematic consistency, were superior. The results suggest that while the AI is strong in certain aspects, a balanced approach considering both human and AI corrections might be beneficial for a more comprehensive evaluation.

Table 6 Average of evaluations done for the corrections

Elements	Average	
	Human	AI
Grammar	4	5
Spelling	4	5
Sentence Structure	4	5
Punctuation	4,5	4
Coherence	3	3,5
Relevance	3	4,5
Essay Structure	4	4
Clarity	3,5	4
Thematic Consistency	4	3
Supporting Evidence	2	5

The notable superiority of ChatGPT in addressing grammar, spelling, and sentence structure can be attributed to its proficiency in language mechanics and rule-based corrections. The AI model, powered by extensive training data, excels in recognizing and rectifying common grammatical errors and spelling mistakes. Additionally, its capacity to comprehend and generate diverse sentence structures contributes to its success in this regard. Moreover, in coherence, relevance, and supporting evidence, ChatGPT showcases its ability to analyze context and provide content aligned with the essay's theme. However, the human advantage in thematic consistency suggests that nuanced understanding and contextual interpretation remain areas where human evaluators excel. The AI's impressive performance in providing supporting evidence could be linked to its vast knowledge base, allowing it to integrate relevant examples and facts seamlessly.

An additional step involved conducting a group interview with 10 randomly selected students from our sample. The interview aimed to gather students' feedback on the grading of their essays. This stage is considered valuable for our research as it allows us to capture perspectives from another angle of thought, in addition to the potential to discover new avenues for further investigation. During the interview, a variety of questions were asked, as indicated in Table 7.

The questions posed to the students have allowed us to draw several conclusions. While they acknowledged the sophistication of the corrections, students expressed difficulties in understanding some of the vocabulary used. The corrections, being linguistically advanced, posed a challenge for students at various proficiency levels, hindering their comprehension. In contrast, when evaluating corrections by professors, students reported a preference for the human touch. They appreciated corrections that were more attuned to their academic level, making the content more accessible and facilitating a better understanding. This suggests that, despite the AI's proficiency, there is a need for a balance that considers the students' comprehension levels and ensures that corrections are tailored to their educational stage.

5 Recommendations

Based on the comprehensive evaluation of essay corrections using both ChatGPT and human experts, coupled with valuable feedback from students, several recommendations emerge. Firstly, while ChatGPT showcases remarkable proficiency in addressing grammatical, spelling, and structural aspects, a balanced approach integrating human expertise is advisable. The nuanced understanding and contextual interpretation exhibited by human evaluators, particularly in maintaining thematic consistency, remain crucial elements that contribute to a holistic assessment. Secondly, in response to students' feedback, it is recommended to refine the language sophistication of ChatGPT's corrections. Striking a balance between linguistic sophistication and adaptability to students' comprehension levels is imperative to enhance overall accessibility. This could involve tailoring the AI's corrections to align with different proficiency levels, ensuring that students at various stages of their academic journey can benefit from the corrections without encountering language barriers. Regular collaboration between AI and human evaluators, incorporating AI advancements while preserving the human touch, is pivotal for a comprehensive and effective essay correction process. Lastly, continuous monitoring, refinement, and adaptation of the analysis sheet based on emerging research and educational standards will contribute to the ongoing improvement of the evaluation methodology. The recommendations can be listed as follows:

- **Balance Human and AI Expertise:**
Advocate for a balanced approach that integrates the strengths of both ChatGPT and human evaluators, acknowledging the nuanced understanding and contextual interpretation that humans bring to the evaluation process.
- **Refine ChatGPT's Language Sophistication:**

Table 7 The interview

Can you share your overall impression of the corrections provided by ChatGPT on your essay?
Did you find the vocabulary used in the corrections to be easily understandable? Why or why not?
How would you compare the corrections made by ChatGPT to those provided by human professors in terms of clarity and relevance to your academic level?
Did you feel that ChatGPT's corrections were too advanced, just right, or not advanced enough for your proficiency level?
Were there aspects of the corrections provided by ChatGPT that you particularly liked or disliked? Why?
How comfortable would you be relying solely on AI-driven corrections for your academic writing, and why?
In comparison to corrections by human professors, do you think ChatGPT's corrections lacked a personal touch? If so, how important is this personal touch to you?

Adjust the language sophistication of ChatGPT's corrections to strike a balance between linguistic complexity and adaptability to students' comprehension levels. Tailor corrections to align with different proficiency levels.

- **Enhance Accessibility:**

Ensure that ChatGPT's corrections are accessible to students at various stages of their academic journey. This involves making language adjustments to prevent language barriers and improve overall comprehension.

- **Collaboration Between AI and Human Evaluators:**

Promote a collaborative environment where AI and human evaluators work in tandem, leveraging the strengths of both entities. This collaboration can contribute to a more comprehensive and effective essay correction process.

- **Continuous Monitoring and Refinement:**

Establish a system for continuous monitoring, refinement, and adaptation of the analysis sheet. Regularly update the criteria based on emerging research, educational standards, and feedback from evaluators to ensure ongoing improvement.

These recommendations aim to create a harmonious integration of AI and human expertise, addressing both the technical and contextual aspects of essay evaluation while maintaining a focus on continuous improvement.

6 Conclusion

In conclusion, the evaluation of essay corrections using both ChatGPT and human expertise has provided valuable insights into the strengths and considerations of each approach. The findings indicate that ChatGPT excels in addressing specific language-related issues such as grammar, spelling, and sentence structure, showcasing its proficiency in rule-based corrections. However, human evaluators maintain an edge in areas requiring complex understanding, particularly in maintaining thematic consistency. It is important to note that these results may not be generalized to contexts involving native English speakers or those with varying levels of language proficiency. Nevertheless, this study suggests its relevance within the framework of Moroccan university settings.

Students' feedback highlights the need for a refined balance in ChatGPT's language sophistication to ensure accessibility and comprehension across various proficiency levels. The recommendations emphasize a collaborative model, leveraging the strengths of both AI and human evaluators, while continuously monitoring and refining the evaluation criteria. This holistic approach aims to create an effective and adaptable essay correction process that enhances overall quality and accessibility. As advancements in AI continue, a dynamic collaboration between technology and human expertise remains essential for optimizing the educational evaluation landscape.

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Author contributions KB conceptualized the research idea and designed the study methodology. They were responsible for data collection, analysis, and interpretation. Additionally, KB drafted the initial manuscript and revised it based on feedback from the co-author. KB also supervised the research project, offering guidance on the research design, methodology, and interpretation of results. They provided oversight on data analysis and ensured adherence to ethical standards in research conduct. AB provided expertise in Artificial Intelligence and contributed to the implementation of AI-based essay correction methods. They played a significant role in analyzing AI-generated corrections and comparing them with human evaluations. AB also contributed to the interpretation of results and provided critical insights into the implications of AI technology in education. He contributed to the design of the study methodology and provided guidance throughout the research process. They played a crucial role in reviewing and validating the data analysis and interpretation. Additionally, both authors provided valuable input in refining the manuscript and ensuring its coherence and clarity.

Data availability Yes it is available; it is attached as excel document with the article and figure.

Declarations

Ethics approval and consent to participate The present study received approval from our institution/our laboratories of Chouaib Doukkali University in accordance with the ethical standards outlined in the 1964 Declaration of Helsinki and its later amendments. All participants were informed about the purpose of the study and provided their written consent to participate. Participation was voluntary, and anonymity was maintained throughout the research process. No personally identifiable information was collected, ensuring confidentiality and privacy of all participants.

Competing interests The authors declare no competing interests.

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